

PROFESSIONAL SUMMARY

Software Engineer and M.S. Computer Science student (GPA: 4.0) with 3+ years of hands-on experience building full-stack and mobile systems using Flutter, Java Spring Boot, and cloud-native technologies. Proven track record taking products from early prototypes to production, owning delivery across backend, frontend, and DevOps. Published researcher in privacy-preserving federated learning for health IoT systems, with peer-reviewed work in *Sensors* (2025) and IEEE CCWC (2026). Seeking full-time software engineering roles where I can contribute across the stack and grow in a high-impact production environment.

TECHNICAL SKILLS

Backend APIs: Java, Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate, RESTful APIs, PostgreSQL, MySQL, Redis, SQL, NoSQL
Frontend: Flutter, Dart, React.js, JavaScript (ES6+), TypeScript, Python, HTML5, CSS3, State Management (Provider, Bloc, Redux), API Integration
Cloud & DevOps: CI/CD Pipelines, Jenkins, GitHub Actions, Google Cloud Platform (GCP), Docker, Kubernetes, Firebase, Cloud Functions, Git
Databases: Relational Databases, Database Design, Schema Modeling, Query Optimization, Indexing, Transactions
Research ML: Federated Learning, Privacy-Preserving ML, Homomorphic Encryption (CKKS), ML Model Deployment

PUBLICATIONS

- [1] R. Kavati, H. Khan, S. S. Pulkaram, and A. Jalooli, "Securing Federated Learning in Health IoT with Edge-Assisted Homomorphic Encryption," in *Proc. IEEE Annual Computing and Communication Workshop and Conference (CCWC)*, 2026. DOI: [10.1109/CCWC62023.2026.11393779](#).
[2] H. Khan, R. Kavati, S. S. Pulkaram, and A. Jalooli, "End-to-End Privacy-Aware Federated Learning for Wearable Health Devices via Encrypted Aggregation in Programmable Networks," *Sensors (MDPI)*, vol. 25, no. 22, p. 7023, Nov. 2025. DOI: [10.3390/s25227023](#). PMID: 41305229.

PROFESSIONAL EXPERIENCE

- Research Assistant** Jul 2024 – Dec 2025
California State University, Dominguez Hills
- Designed privacy-preserving federated learning pipelines using CKKS homomorphic encryption for health IoT systems.
 - Implemented secure model aggregation with in-network computing, offloading encryption overhead from wearable devices to edge nodes.
 - Evaluated EAH-FL on real-world healthcare datasets across 10–60 clients, achieving F1-score > 0.93 and PDR > 0.95.
 - Published peer-reviewed findings in *Sensors* (2025) and IEEE CCWC (2026); see Publications section above.
- Full Stack Engineer** May 2022 – Aug 2024
Floryn Health Pvt. Ltd., Hyderabad, India
- Owned end-to-end design of mobile and backend systems serving healthcare workflows across a 10+ engineer team.
 - Built secure RESTful APIs, JWT authentication, and microservices using Spring Boot and PostgreSQL for thousands of users.
 - Led architectural decisions and delivery timelines, mentoring engineers across three cross-functional teams.
 - Built CI/CD pipelines using Jenkins and GCP, improving deployment frequency by ~3× over manual workflows.
- Founding Engineer** Mar 2021 – Apr 2022
HelloDukaan, Hyderabad, India
- Led full engineering for a consumer e-commerce platform, taking the product from zero to production as the sole founding engineer.
 - Architected cross-platform mobile apps in Flutter, delivering consistent performance across Android and iOS from a single codebase.
 - Built Jenkins CI/CD pipelines, cutting build-to-release cycle time across development, staging, and production environments.
 - Made architectural trade-offs under startup constraints, maintaining reliability as user load scaled post-launch.
 - Collaborated with product and business stakeholders to translate evolving requirements into technical designs and deliverables.

PROJECTS

- SuperQuick (Quick Commerce Platform)** Flutter, Firebase (Firestore, Cloud Functions, FCM)
<https://superquick.in>
- Built a hyperlocal quick-commerce platform solo, covering ordering, fulfillment, delivery assignment, and live order tracking.
 - Developed four Flutter apps Customer, Delivery, Store, and Dark Store Inventory with shared architecture and reusable components.
 - Implemented real-time order state sync using Firestore listeners and optimized data models for low-latency updates end-to-end.
 - Built dark store inventory workflows: stock updates, availability toggles, and pricing changes with instant cross-client propagation.
 - Set up FCM push notification flows for order events, rider assignment, and delivery milestones.
- Brain Tumor Detection (CNN)** Python, CNN, IBM Watson, Node-RED
- Built a CNN model to classify MRI scans and detect brain tumors on labeled medical imaging datasets.
 - Deployed the inference pipeline via IBM Watson and Node-RED with a user-facing interface for image uploads and predictions.

EDUCATION

M.S. Computer Science – California State University, Dominguez Hills Expected Dec 2026 (GPA: 4.0)
B.Tech Computer Science – Sree Dattha Institute of Engineering and Science 2020